SEVARO AT RUTGERS HEALTH HACK

Challenge 2 Handbook

What the problem is, the issues to plan for, how your tool will be tested, and what a strong answer does.

THE SHORT VERSION

Sevaro's neurologists look after patients at many hospitals over video. Sometimes a hospital suddenly gets a rush of calls, often stroke cases, just as doctors are about to go home. There may not be enough doctors on shift, or not enough who are approved to work at that hospital, and a case ends up waiting with nobody free to take it. Minutes matter in a stroke, so this has to be caught and fixed fast.

How it's handled today. Calls are sent to doctors automatically, and a live screen shows what each doctor is doing right now. When a rush hits, it's up to a person to spot it, work out by hand who could help, and get hold of them. That works on a quiet day and slips on a busy one.

Teams build a tool that **notices this early, decides what to do about it, and gets it done**, for example by asking an off-duty doctor to work a few extra hours, with a person approving every offer.

CHAPTER 1

The situation

Sevaro's neurologists cover more than 200 hospitals remotely, by phone and video. Calls, video and each doctor's status run through **Synapse**, Sevaro's platform. Each request from a hospital is a **consult**. **Stroke** consults are the most urgent: every minute counts.

Picture a normal evening. Calls come in and are sent automatically to doctors who are on shift and approved at that hospital. Then a bump: one hospital suddenly sends several stroke consults in half an hour, just as two doctors are about to finish their shift.

- A stroke consult is waiting and **nobody approved at that hospital is free**.
- The doctors about to leave still have **unfinished notes**.
- In an hour **fewer doctors will be on**, so it gets worse, not better.

The fixes exist: move a consult to another doctor, escalate to the clinical lead, or **call in an off-duty doctor** for a few hours. The hard part is knowing quickly which fix is right, and who is allowed to help.

Who's involved

ON SHIFT

Doctors working now

Each juggles several consults at once: on a call for one, waiting on a scan for another, writing up a third.

OFF SHIFT

Doctors not working

Resting or between shifts. Some have said they'd take a few extra hours if asked.

OPERATIONS

The coordinator

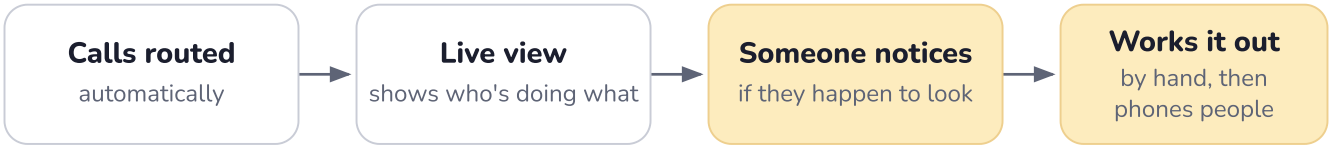
Watches the whole network, moves consults between doctors, and proposes extra help.

CLINICAL

The clinical lead

Approves or rejects extra-shift offers and authorizes escalations.

How it's handled today



The amber steps are manual. That's where things slip.

Build a dashboard, but not a copy of the live screen. Sevaro already has a screen showing what's happening now, and a tool that only does that will score poorly. The value is in what happens after a problem is spotted: deciding what to do, and getting it done.

CHAPTER 2

The issues to plan for

Whether a waiting stroke patient gets a doctor in time depends on many things at once. **Before you build, list them yourselves**, then check your list against these. Each comes with an example and what to anticipate, because the data you're tested on won't be the data you practiced on.

1. A sudden rush at one hospital

Demand is uneven. One hospital can send far more consults in an hour than it normally does.

Example: A hospital that usually sends 2 consults an hour sends 7 in 45 minutes.

Anticipate: The rush may hit a different hospital, at a different time, or two at once. Compare each hospital with its own normal level, not a fixed number.

2. Doctors leaving while it's busy

Shifts end on schedule whether or not it's quiet. Cover drops, and leaving doctors may still have unfinished notes.

Example: Three doctors finish at 8pm with five open consults between them, just as a rush starts.

Anticipate: Look ahead at least 90 minutes. Expect shift times, and who starts next, to change between datasets.

3. Free, but not approved there

A doctor can only take consults from hospitals where they're credentialed. Being free isn't enough.

Example: Three doctors are free, but none of them is approved at the hospital with the waiting stroke patient.

Anticipate: Credentialing will be different in new data. Never assume any doctor can cover any hospital.

4. "Available" isn't always available

Busy or free is worked out from signals, and they can disagree. The status a doctor sets is only one clue.

Example: A doctor's status says Available, but their camera is on with no Synapse call. They're probably on another video call.

Anticipate: Expect signals that conflict, the desktop helper app (which reports whether a doctor's camera is on) going quiet mid-day, and a doctor who goes unavailable while a consult is still theirs.

5. A consult stuck at one step

Every step of a consult has a time limit, tighter for strokes. Something waiting too long is a problem even if nobody has noticed.

Example: A stroke consult has had scans back for 14 minutes and the video hasn't started. The critical limit is 12 minutes.

Anticipate: Measure time from when each step started, using when things happened, not when records arrived.

6. Nobody can take the call

When nobody qualifies, the automatic routing fails and retries every 5 minutes. The hospital is left waiting.

Example: A stroke consult fails routing twice, and the hospital has now waited 15 minutes.

Anticipate: Expect more than one stuck consult at once. Rank them, and say which to deal with first and why.

7. The same patient, twice

Records can arrive late or twice. A hospital that gets no answer may also call again about the same consult.

Example: A consult comes in at 6:05, nobody picks up, and the hospital calls again about that same consult at 6:12. That's one patient who has waited 7 minutes, not a new one.

Anticipate: Don't count a repeat as new demand, and don't restart its clock; it should move up the list. But different consults at the same hospital are different patients. Never merge them on a guess, or a real rush disappears.

8. Who can be called in

An off-duty doctor can only be asked if they pass every rule: willing, approved at a short hospital, rested, under their hours for the last 7 days, no clash, long enough to be worth it.

Example: Three off-duty doctors look possible. One finished a shift 8 hours ago, one has already had 3 offers this week, and one needs so long to get online that the offer is worth under 2 hours.

Anticipate: In new data a different doctor may be the one, or nobody may qualify. Then the answer is to escalate, not to bend a rule.

9. The extra doctor gets swamped

Routing favors doctors with the fewest consults. Someone joining mid-evening has zero, so they'd get nearly every call everywhere they're approved.

Example: A doctor joins to cover one short hospital and gets the next six calls from five different hospitals.

Anticipate: Show how your plan keeps the new doctor on the hospitals that need help, and what happens to everyone else's calls.

10. Approvals and safety

A person approves every offer. Messages carry no patient details. Offers expire, and two doctors might accept the same one.

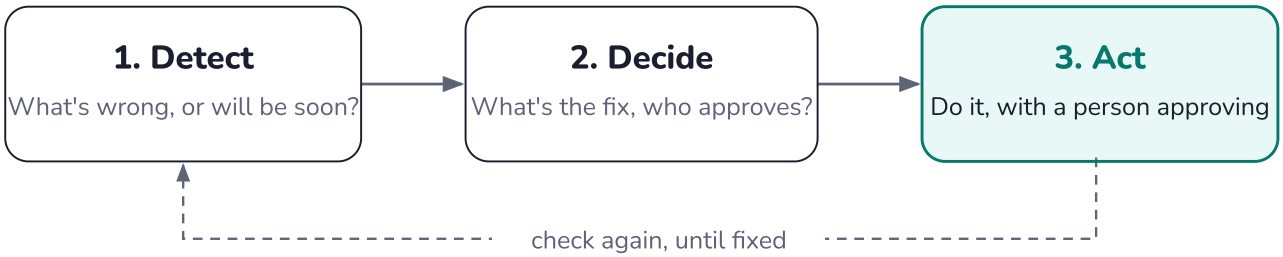
Example: Two doctors accept the same offer within a minute of each other.

Anticipate: Expect replies that come late, never come, or clash. Your tool should handle each one cleanly and keep a record.

CHAPTER 3

What to build

A tool with three connected parts. All three are required.



- **Detect.** What needs attention now, and what will in the next 60 to 90 minutes? Consults stuck too long, a hospital far busier than normal, too few approved doctors once shifts end, signals that don't add up.
- **Decide.** What's the right fix, and who has to approve it? Move a consult, escalate, or offer an off-duty doctor a short extra shift. For an extra shift, show every off-duty doctor and why each one can or can't be asked.
- **Act.** Do it and follow it through to the end: get approval, send the offer, track accept, decline or no reply, update the picture, and stop asking once the gap is filled.

A person stays in charge. Only small, safe steps (like an internal alert to the coordinator) may happen automatically. Any message to an off-duty doctor needs a person to approve it.

You'll need a dashboard

People have to see the problems and act on them, so build a dashboard on top of your engine. It should let the **coordinator** see what needs attention and propose a fix, the **clinical lead** approve or reject it, and show a stand-in **phone screen** where the off-duty doctor accepts or declines. How it looks is up to you.

What a strong one looks like

At 6:20pm the coordinator opens it and quickly knows **the one thing to act on first**, **why**, **what to do**, and **who approves it**. Soon after, the fix is underway, and the record shows exactly what happened.

PERSON	WHEN IT WORKS, THEY CAN...
Coordinator	See every problem ranked by urgency with a plain reason, see which hospitals will be short soon, and act without working anything out by hand.
Clinical lead	See what puts stroke care at risk, with the proposed fix and the reasoning, and approve or reject in one step.
Off-duty doctor	Get a clear offer on their phone (which hospitals, which hours, when, and why them) and accept or decline. Nothing else.

CHAPTER 4

The test: new data

During your demo we will give you one CSV file that looks just like event_stream.csv: the same 8 columns (event_id, event_time, ingested_at, consult_id, event_type, physician_id, hospital_id, description). **This is the file we test you with:** you drop it into your tool. Your tool has to add it to the day you already have, run it, and on its own sort out the problems: **what is going on, where, why it matters, and the fix it suggests.**

How it compares with the kit. The file is comparable to event_stream.csv: the same columns, the same doctors and hospitals, the same kinds of events. What's different is what happens in it: new records that move the day forward and change who is free, what is stuck and what needs doing next.

So the heart of this is an engine. It should read whatever records it's given, rebuild what is happening from them, and work out the problems and fixes by itself. Your dashboard shows what the engine found and lets people act on it.

- Build for data you haven't seen. A tool tuned to the sample day will break when the names, times and hospitals change.
- Real data is messy in all the ways described in Chapter 2. Your engine should cope with every one of them, every time.

For each problem, your tool should say

QUESTION	FOR EXAMPLE
What is going on?	A stroke consult is waiting and nobody can take it.
Where?	Which hospital, which consult, which doctor.
Why does it matter?	How long it has waited against its limit, and whether it gets worse when shifts end.
What should we do?	The suggested fix, who has to approve it, and why that fix over the others.

What the brief asks you to show

1. Rebuild the picture at 6:20pm from the records, handling late and duplicate ones, and match the reference snapshot.
2. Show what each doctor is doing and where each consult is, separately, with time waited against each limit.
3. Find and rank what needs attention, and explain why each item was flagged.
4. Look ahead at least 90 minutes, accounting for shift ends and normal demand.
5. Recommend a fix for the top problem. For an extra shift, show every off-duty doctor and which rule each passed or failed.
6. Run the whole loop: propose, approve, send (pretend), accept or decline, update the picture.
7. Show how you stop a newly joined doctor from being swamped.
8. Update without a restart when that new file is added during your demo.

Easy ways to lose ground

- Only showing what's happening now.
- Trusting the status a doctor sets for themselves.
- Ignoring who is approved at which hospital.
- Reading records in the order they arrived instead of when they happened.

- Sending an offer without approval, or with patient details in it.
- Needing a restart to load new data.

CHAPTER 5

The rules, in short

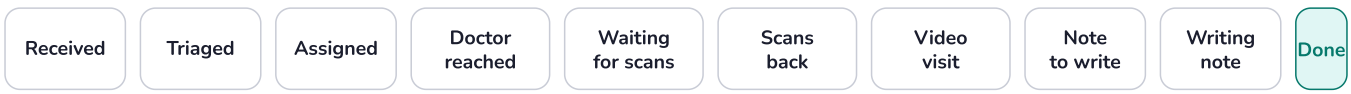
All numbers here are made up for the hackathon and are not Sevaro's real settings.

Busy or free

Doctors don't press a "busy" button. It has to be worked out from signals:

SIGNAL	READ IT AS
On a Synapse phone call or video	Busy.
Camera on, no Synapse video, status not "In Meeting"	Probably on a video call somewhere else. Busy, and flag it.
Camera on, status "In Meeting"	An internal meeting that can be interrupted. Available.
Desktop helper app (reports the camera) gone quiet	Looks available, but nobody can be sure. Point out the risk.
Status they set themselves	Often says Available even when they're busy. Only one clue.

Time limits for each step



Not every consult goes through every step. A phone-only consult skips the video.

WAITING AT THIS STEP	APPLIES TO	WARNING	CRITICAL
Triaged, no doctor assigned yet	All	5 min	10 min
Assigned, but the hospital hasn't reached the doctor yet	Stroke	3 min	8 min
Waiting for scans	Stroke	12 min	20 min
Scans back, video not started	Stroke	6 min	12 min
Note not finished after the call or video	Stroke	30 min	45 min
Note not finished after the call or video	General	45 min	90 min
Unfinished notes within an hour of shift end	All	Any	2 or more

How calls are routed

1. Only doctors who are on shift and approved at that hospital.
2. Skip anyone busy, anyone whose phone is already ringing, and anyone who hung up less than 2 minutes ago.
3. Doctors with very few consults this shift go first (6 or fewer on days, 4 or fewer at night).
4. Otherwise, the fewest consults this shift. Ties go to whoever answered longest ago.
5. If nobody qualifies, it fails and tries again every 5 minutes.

The catch. A doctor who joins mid-evening starts with 0 consults, so these rules send them nearly every call at every hospital they're approved for, not just the hospitals that needed help. They get swamped and the problem just moves. Show how you prevent that.

When a hospital is short

Each hospital needs a minimum number of available, approved doctors at each time of day. It's short when it has fewer, now or at a time coming up (like when shifts end). If its last hour was

busier than its normal busy level for that hour of the day, that's a surge, and the minimum goes up by one. Look at least 90 minutes ahead.

Calling in an off-duty doctor

They can only be asked if **all** of these are true: they've said they're willing; they're approved at a hospital that is short; 10 hours have passed since their last shift (at the time the extra shift starts); their last 7 days plus the extra hours stay at 60 or under; it doesn't clash with a scheduled shift; and it's at least 2 hours long. The time they need to get online counts against what the offer is worth.

And for every offer: the clinical lead approves it first, no patient details in the message, one open offer per doctor and no more than 3 a week, it expires after 10 minutes, decide what happens if two doctors accept the same offer, and everything is logged (who was considered, who wasn't and why, who approved, what happened).

Records that arrive late, twice, or again

Every record has two times: when it happened and when it arrived. Records can arrive late, or the same record can arrive twice. Put them in the order they happened and count each one once.

There's a real-world duplicate too: a hospital calls about a consult, nobody picks up, and they call again about the same one. That's the same patient. Don't count it twice or restart its clock; it has been waiting since the first call and should move up the list. Different consults are different patients.

CHAPTER 6

The data

One day: **Tuesday 17 March 2026, 7am to 6:20pm** Eastern. Treat **6:20pm as "now"**, not the computer's clock. 12 hospitals across four time zones, 8 doctors on shift, 6 off. Everything is made up. Don't use real patient, doctor or hospital data of any kind.



The window teams must look ahead into: at least 90 minutes.

FILE	WHAT'S IN IT
event_stream.csv	About 1,160 records of what happened during the day. The source of truth.
hospitals.csv	The 12 hospitals and their time zones.
physicians_on_shift.csv	The 8 doctors working now, with shift times.
physicians_off_shift_roster.csv	The 6 off-duty doctors: willing or not, last and next shift, hours in the last 7 days, time to get online.
credentialing_matrix.csv	Which doctor is approved at which hospital.
baseline_hourly_volume.csv	Normal consults per hospital per hour, and the "normal busy" level.
coverage_requirements.csv	Minimum available doctors each hospital needs, by time of day.
snapshot_*_1820.csv	The correct picture at 6:20pm, to check your work against.

What's hidden in it

The brief lists the situations a good tool should catch. Some only show up if you replay the records correctly or look ahead:

- A stroke consult nobody approved can take.
- A hospital far busier than normal.
- Two doctors near the end of their shift with open work, during that rush.
- A doctor who goes unavailable while a stroke consult is still theirs.
- A camera on with no Synapse call, and a doctor whose helper app has gone quiet.
- A record that arrives late, and one that appears twice.
- Uneven workload caused by where doctors are approved, not effort.
- Exactly one off-duty doctor who can be asked. Several others look close but each fails one rule.

CHAPTER 7

Going further

Not required, but worth thinking about. Pick one or two that matter to your idea and factor them in.

- **Cost:** what an extra shift costs, compared with what a late stroke consult costs.
- **Fitting into Sevaro's systems:** where the real data would come from, and how offers would really be sent.
- **Patient privacy:** HIPAA, the US health privacy law, and what can and can't go in a message.
- **Licensing:** doctors need a license in the hospital's state as well as approval at that hospital.
- **Doctor fatigue and fairness:** rest, hours, and not always asking the same person.
- **Alert overload:** too many warnings and people stop reading them.
- **When data stops arriving:** how the tool shows that it's out of date.

CHAPTER 8

Words to know

Consult

One hospital's request for a neurologist for one patient.

Stroke consult

The most urgent consult type, with the tightest time limits.

Synapse

Sevaro's platform for calls, video and status. Its records are the data.

Triage

The first check of a new consult.

Credentialed

Approved to treat patients at a particular hospital.

Licensed

Allowed to practice medicine in a particular state. Real-world only; the data folds this into credentialing.

Split shift

An off-duty doctor coming online for a few extra hours to cover a gap.

Coverage gap

A hospital with fewer available, approved doctors than it needs.

Surge

A hospital getting more consults in the last hour than its normal busy level for that hour of the day.

p90

The "normal busy" level for a hospital at a given hour of the day. Over the past 12 weeks, that hour was at or below it 9 times out of 10.

Escalate

Hand a problem up to the clinical lead because the normal fix isn't available.

HIPAA

The US law protecting patients' health information.